Wall time per step: Splat simulation, 1M (964,763 cells)

Seconds per step. Colour is scaled per step and size: fastest framework 0, slowest 1, on square−root seconds.

	bixverse mtx	bixverse h5ad	bixverse GPU*	bixverse MLX*	Scanpy mtx	Scanpy h5ad	Scanpy dask	scrapper mtx	scrapper h5ad	scrapper HDF5	Seurat std mtx	Seurat std h5ad	Seurat BPCells	CyteArc mtx	CyteArc h5ad
load	90	57	49	42	OOM	OOM	39	OOM	OOM	405	OOM	OOM	129	303	67
qc	0.5	0.4	0.4	0.4	–	–	7.2	–	–	12	–	–	8.6	0.1	0.1
normalise	n/a	n/a	n/a	n/a	–	–	0.0	–	–	0.0	–	–	4.1	27	28
hvg	2.0	2.2	1.6	1.5	–	–	8.2	–	–	38	–	–	25	13	14
pca	4.8	5.2	2.6	2.1	–	–	18	–	–	144	–	–	463	50	54
harmony	3.2	3.0	2.0	1.2	–	–	4.2	–	–	166	–	–	175	163	148
knn_snn	29	30	15	13	–	–	40	–	–	87	–	–	95	364	393
cluster	8.2	8.2	6.7	6.5	–	–	47	–	–	269	–	–	343	19	20
cluster_fast	46	47	7.9	6.3	n/a	n/a	n/a	–	–	545	n/a	n/a	n/a	n/a	n/a
umap	80	80	11	9.7	–	–	555	–	–	362	–	–	546	863	869
dge	4.1	4.2	3.6	3.3	–	–	–	–	–	TIMEOUT	–	–	518	24	24
tsne_default	654	628	562	544	–	–	TIMEOUT	n/a	n/a	n/a	–	–	TIMEOUT	n/a	n/a
tsne_fft	194	197	30	46	–	–	–	n/a	n/a	n/a	–	–	–	n/a	n/a
tsne_quick	134	134	84	73	n/a	n/a	n/a	–	–	–	n/a	n/a	n/a	n/a	n/a
total (no tSNE)	222	190	91	80	failed	failed	719	failed	failed	failed	failed	failed	2,305	1,826	1,617

Black: n/a (step not in that framework), OOM or TIMEOUT where the run died, − not reached.
Each step is scaled over the runs that finished it: at 1M the bixverse CPU tSNE is red because the others timed out.
Scanpy dask has no DGE step. bixverse GPU and MLX run natively without a memory cap.
bixverse CPU tSNE builds its kNN with IVF from 500k (kmknn at 100k). (qd) steps are not in the total.